

Sitara Project B'd (Rev.A)

[1] Index

Rev	Date	Designer	Description
A	25.12.20	Ganghyeok Lim	Create design project

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- #6 Sitara DDR
- #7 Sitara I/O Bank 1/2/3

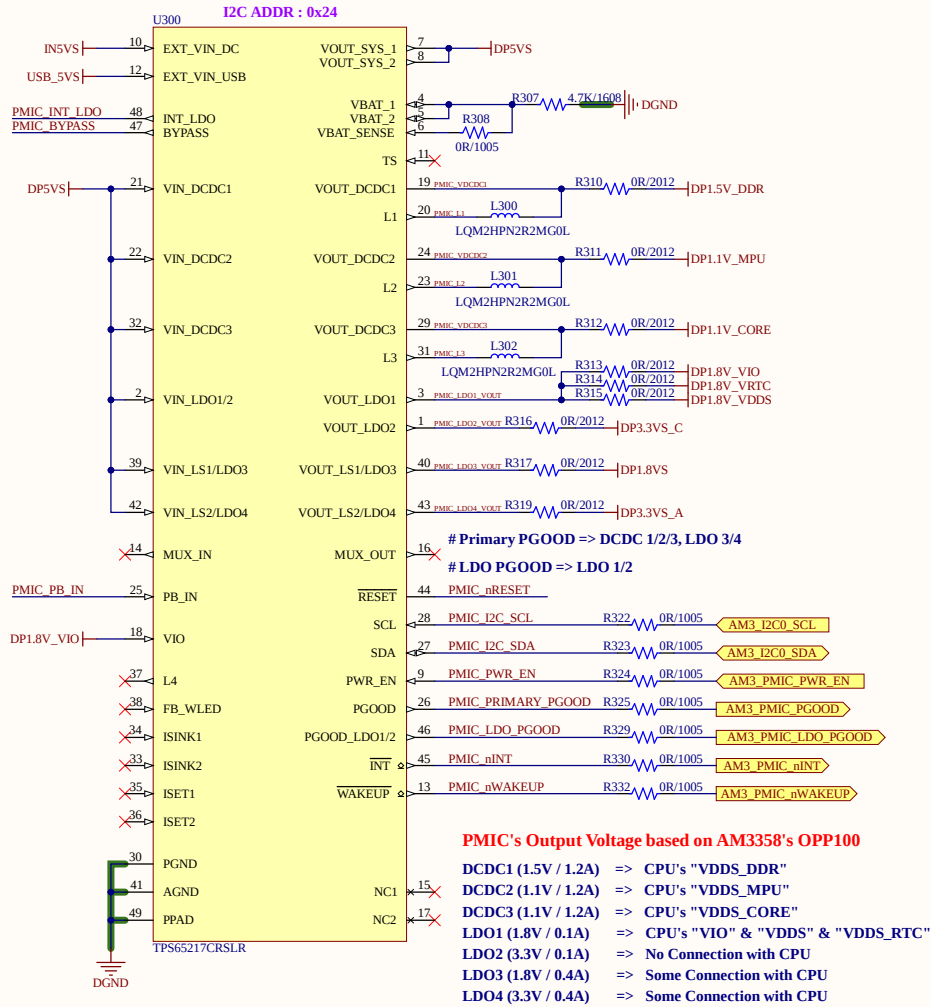
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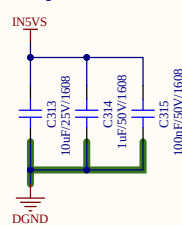
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A	Sitara Project B'd (Rev.A) [2] Overview																								
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D	<table><tr><td>Title</td><td colspan="2">Sitara_Project_Rev_A.PrjPcb</td><td>Rev</td><td>*</td></tr><tr><td>Doc</td><td colspan="2">Overview.SchDoc</td><td colspan="2" rowspan="3"></td></tr><tr><td>Sheet # 2 of 10</td><td>Author</td><td>*</td></tr><tr><td>Date</td><td colspan="2">2025-12-28</td></tr><tr><td colspan="5">This document is copyright of ArkX and shall not be revealed, produced, copied, in whole or in part, nor used for any purpose other than submitted.</td></tr></table>				Title	Sitara_Project_Rev_A.PrjPcb		Rev	*	Doc	Overview.SchDoc				Sheet # 2 of 10	Author	*	Date	2025-12-28		This document is copyright of ArkX and shall not be revealed, produced, copied, in whole or in part, nor used for any purpose other than submitted.				
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	1	2	3	4																					

[3] Power

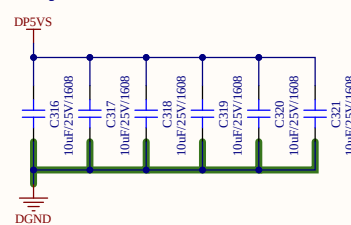


Decoupling Capacitors

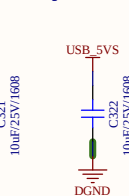
Decap for IN5VS



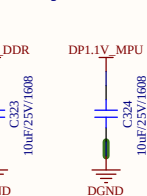
Decap for DP5VS



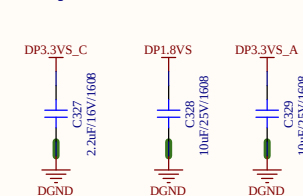
Decap for USB_5VS



Decap for DCDC 1/2/3

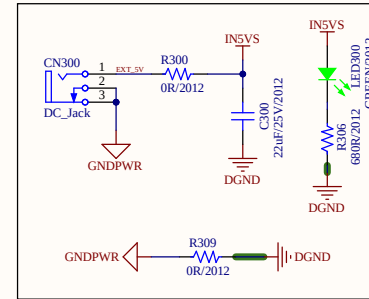


Decap for LDO 1~4

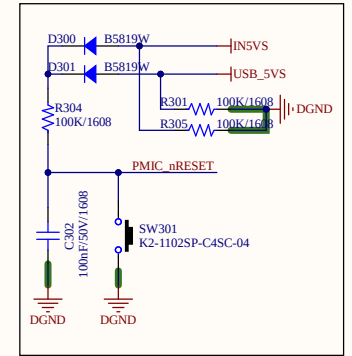
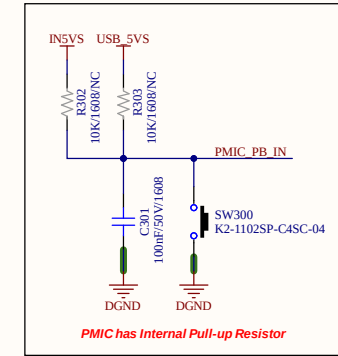


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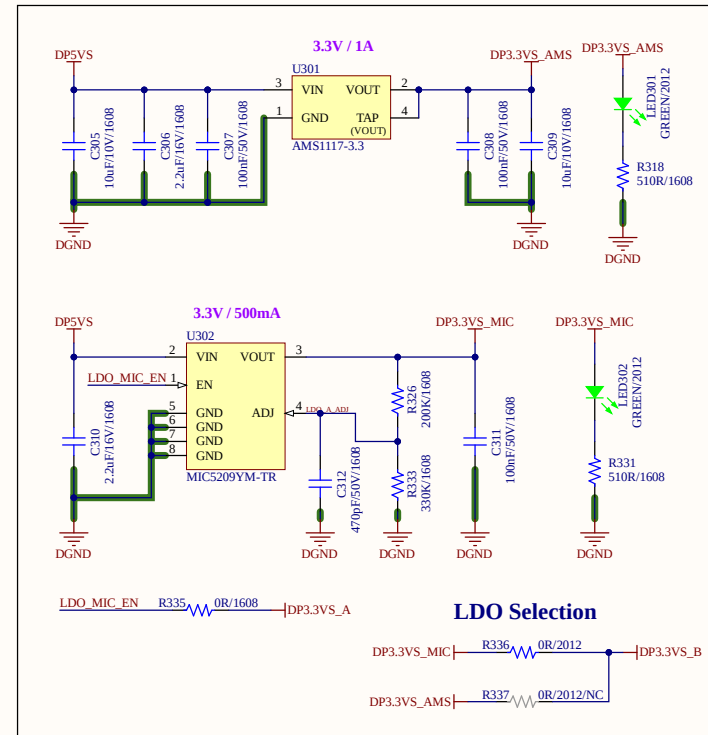
Power Connector (5V)



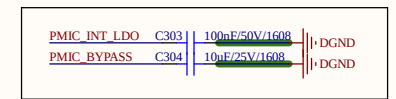
PMIC Power-On Button PMIC Reset Button



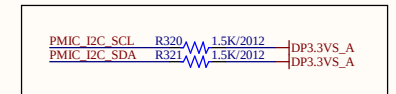
LDO (3.3V / 500mA)



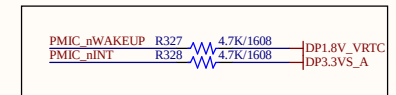
INT LDO / BYPASS Pin



I2C Pull-up Resistor



nWAKE-UP / nINT Pull-up

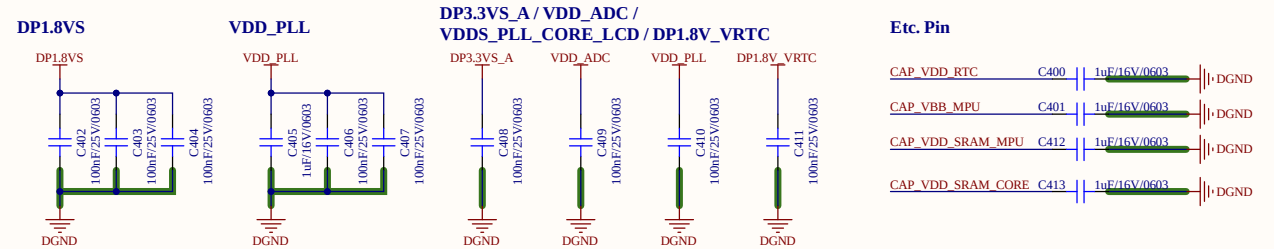
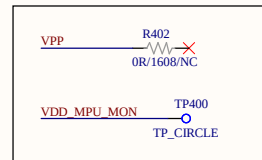
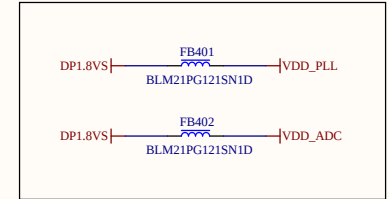
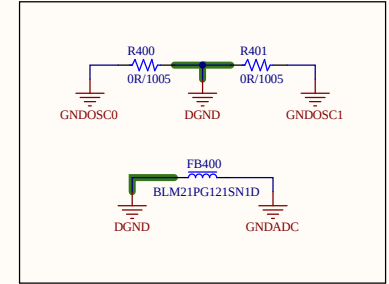
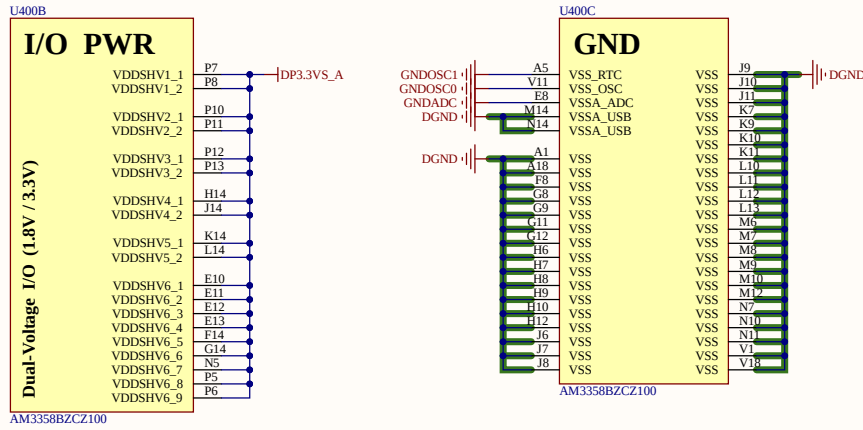


PMIC Power LED



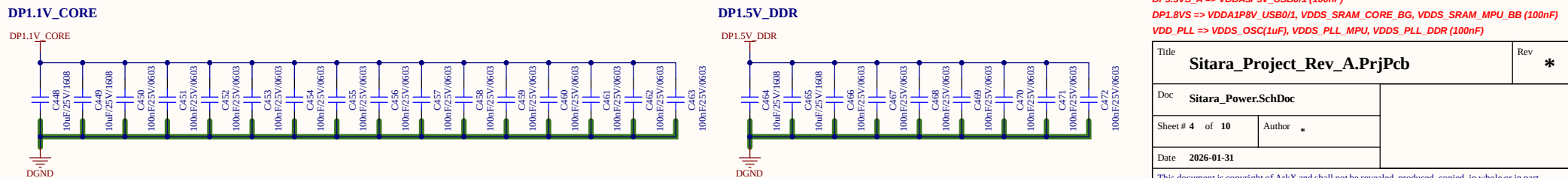
Rail	Output [V]	Sequence (Strobe)
DCDC1	1.5 [V] / 1.2 [A]	1
DCDC2	1.1 [V] / 1.2 [A]	5
DCDC3	1.1 [V] / 1.2 [A]	5
LDO1	1.8 [V] / 0.1 [A]	15
LDO2	3.3 [V] / 0.1 [A]	3
LS1 or LDO3	1.8 [V] / 0.4 [A]	2
LS2 or LDO4	3.3 [V] / 0.4 [A]	4

Place Decaps Closely to Each Pin of PMIC



The diagrams illustrate the decoupling capacitor placement for three different power rails:

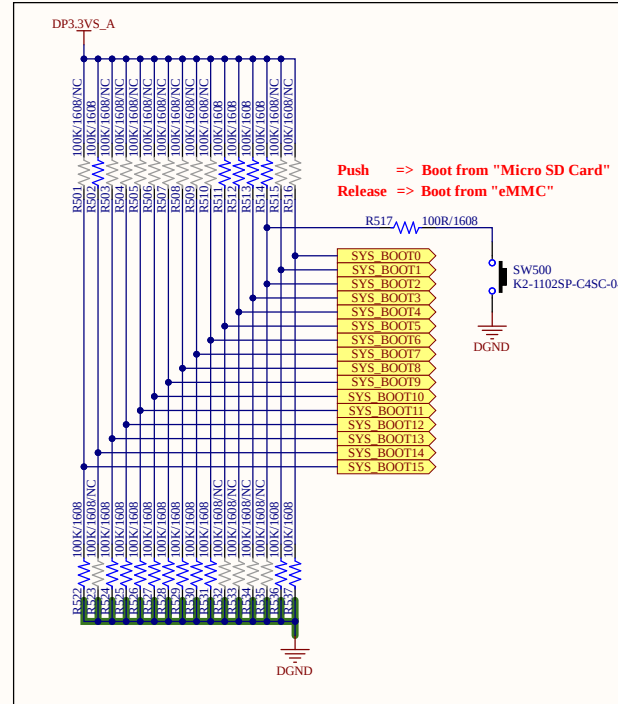
- DP1.1V_MPU:** A power rail labeled DP1.1V_MPU is connected to a series of capacitors C414 through C419. Each capacitor is connected to a common ground (DGND). The capacitors are labeled with their values: C414 (100nF/25V/1608), C415 (100nF/25V/0603), C416 (100nF/25V/0603), C417 (100nF/25V/0603), C418 (100nF/25V/0603), and C419 (100nF/25V/0603).
- DP1.8V_VDDS:** A power rail labeled DP1.8V_VDDS is connected to a series of capacitors C420 through C426. Each capacitor is connected to a common ground (DGND). The capacitors are labeled with their values: C420 (100nF/25V/1608), C421 (100nF/25V/0603), C422 (100nF/25V/0603), C423 (100nF/25V/0603), C424 (100nF/25V/0603), C425 (100nF/25V/0603), and C426 (100nF/25V/0603).
- VDDSHV1~6:** A power rail labeled DP3.3VS_A is connected to a series of capacitors C427 through C447. Each capacitor is connected to a common ground (DGND). The capacitors are labeled with their values: C427 (100nF/25V/1608), C428 (100nF/25V/0603), C429 (100nF/25V/0603), C430 (100nF/25V/0603), C431 (100nF/25V/0603), C432 (100nF/25V/0603), C433 (100nF/25V/0603), C434 (100nF/25V/0603), C435 (100nF/25V/0603), C436 (100nF/25V/0603), C437 (100nF/25V/0603), C438 (100nF/25V/0603), C439 (100nF/25V/0603), C440 (100nF/25V/0603), C441 (100nF/25V/0603), C442 (100nF/25V/0603), C443 (100nF/25V/0603), C444 (100nF/25V/0603), C445 (100nF/25V/0603), C446 (100nF/25V/0603), and C447 (100nF/25V/0603).



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Boot Config

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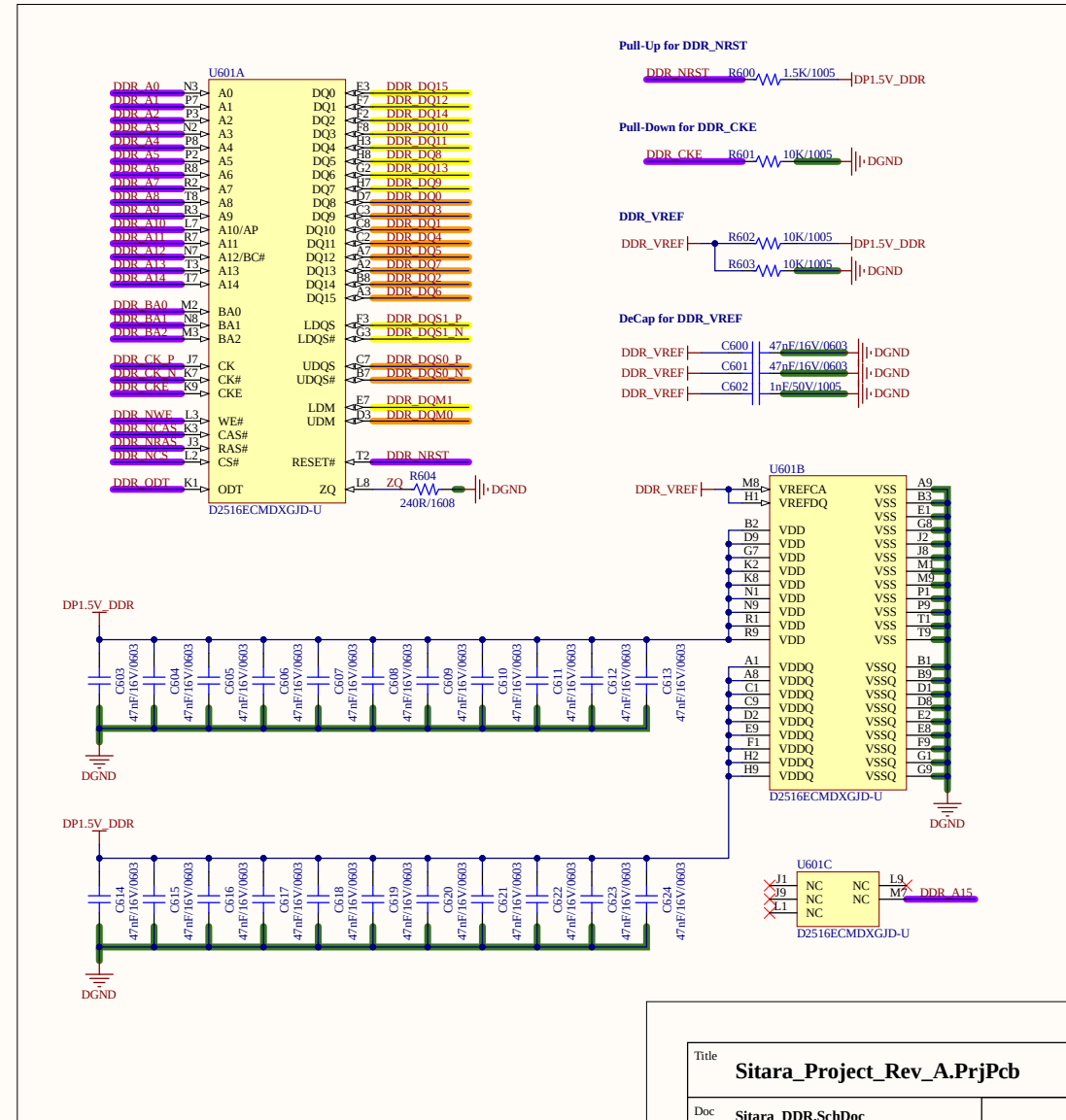
RTC_KALDO_nEN R538 10K/1005 || DGND

The top diagram shows the connection of an SX2M24.000M20F30T1NN oscillator. The VDD pin (pin 4) is connected to VDD_PLL, and the EN pin (pin 1) is connected to a 100nF/25V/0603 capacitor (C501) to GNDOSC0. The OUT pin (pin 3) is connected to EXT_CLK_24MHZ, and the GND pin (pin 2) is connected to GNDOSC0.

The bottom diagram shows the connection of an SX2M32.768KHZ20F30T1NN oscillator. The VDD pin (pin 4) is connected to DP1.8V_VRTC, and the EN pin (pin 1) is connected to a 100nF/25V/0603 capacitor (C505) to GNDOSC1. The OUT pin (pin 3) is connected to EXT_CLK_32.768KHZ, and the GND pin (pin 2) is connected to GNDOSC1.

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DDR3L Memory



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The schematic diagram illustrates the electrical connections for the MTF32GAZAHQD-WT board, featuring two main components: U703A and U702B.

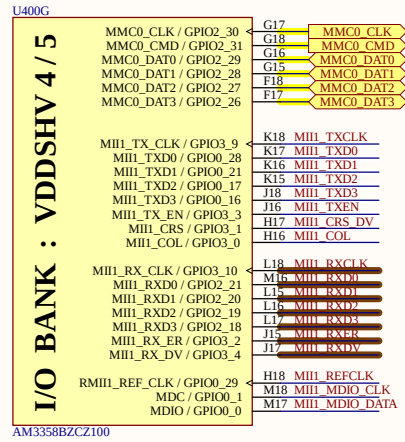
U703A (MTF32GAZAHQD-WT) Connections:

- Power and Ground:** VCC1, VCC2, VCC3, VCC4 are connected to DP1.8VS_eMMC. VCCQ1, VCCQ2, VCCQ3, VCCQ4, VCCQ5 are connected to DP1.8VS_eMMC. VSS1, VSS2, VSS3, VSS4, VSS5, VSS6, VSSQ1, VSSQ2, VSSQ3, VSSQ4, VSSQ5 are connected to DGND.
- Control Signals:** CLK, RST, CMD are connected to DP1.8VS_eMMC. DAT0 through DAT7 are connected to MMCI_0 through MMCI_7.
- Serial Resistor Placement:** A callout indicates "Place Serial Resistors Close to MPU". Resistors R717, R718, R719, R720, R721, R722, R723, R724, R725, R726, R727, R728, R729, R730, R731, R732, R733, R734, R735, R736, R737, R738, R739, R740, R741, R742, R743, R744, R745, R746, R747, R748, R749, R750, R751, R752, R753, R754, R755, R756, R757, R758, R759, R760, R761, R762, R763, R764, R765, R766, R767, R768, R769, R770, R771, R772, R773, R774, R775, R776, R777, R778, R779, R780, R781, R782, R783, R784, R785, R786, R787, R788, R789, R790, R791, R792, R793, R794, R795, R796, R797, R798, R799, R800, R801, R802, R803, R804, R805, R806, R807, R808, R809, R810, R811, R812, R813, R814, R815, R816, R817, R818, R819, R820, R821, R822, R823, R824, R825, R826, R827, R828, R829, R830, R831, R832, R833, R834, R835, R836, R837, R838, R839, R840, R841, R842, R843, R844, R845, R846, R847, R848, R849, R850, R851, R852, R853, R854, R855, R856, R857, R858, R859, R860, R861, R862, R863, R864, R865, R866, R867, R868, R869, R870, R871, R872, R873, R874, R875, R876, R877, R878, R879, R880, R881, R882, R883, R884, R885, R886, R887, R888, R889, R890, R891, R892, R893, R894, R895, R896, R897, R898, R899, R900, R901, R902, R903, R904, R905, R906, R907, R908, R909, R910, R911, R912, R913, R914, R915, R916, R917, R918, R919, R920, R921, R922, R923, R924, R925, R926, R927, R928, R929, R930, R931, R932, R933, R934, R935, R936, R937, R938, R939, R940, R941, R942, R943, R944, R945, R946, R947, R948, R949, R950, R951, R952, R953, R954, R955, R956, R957, R958, R959, R960, R961, R962, R963, R964, R965, R966, R967, R968, R969, R970, R971, R972, R973, R974, R975, R976, R977, R978, R979, R980, R981, R982, R983, R984, R985, R986, R987, R988, R989, R990, R991, R992, R993, R994, R995, R996, R997, R998, R999, R1000, R1001, R1002, R1003, R1004, R1005, R1006, R1007, R1008, R1009, R1010, R1011, R1012, R1013, R1014, R1015, R1016, R1017, R1018, R1019, R1020, R1021, R1022, R1023, R1024, R1025, R1026, R1027, R1028, R1029, R1030, R1031, R1032, R1033, R1034, R1035, R1036, R1037, R1038, R1039, R1040, R1041, R1042, R1043, R1044, R1045, R1046, R1047, R1048, R1049, R1050, R1051, R1052, R1053, R1054, R1055, R1056, R1057, R1058, R1059, R1060, R1061, R1062, R1063, R1064, R1065, R1066, R1067, R1068, R1069, R1070, R1071, R1072, R1073, R1074, R1075, R1076, R1077, R1078, R1079, R1080, R1081, R1082, R1083, R1084, R1085, R1086, R1087, R1088, R1089, R1090, R1091, R1092, R1093, R1094, R1095, R1096, R1097, R1098, R1099, R1100, R1101, R1102, R1103, R1104, R1105, R1106, R1107, R1108, R1109, R1110, R1111, R1112, R1113, R1114, R1115, R1116, R1117, R1118, R1119, R1120, R1121, R1122, R1123, R1124, R1125, R1126, R1127, R1128, R1129, R1130, R1131, R1132, R1133, R1134, R1135, R1136, R1137, R1138, R1139, R1140, R1141, R1142, R1143, R1144, R1145, R1146, R1147, R1148, R1149, R1150, R1151, R1152, R1153, R1154, R1155, R1156, R1157, R1158, R1159, R1160, R1161, R1162, R1163, R1164, R1165, R1166, R1167, R1168, R1169, R1170, R1171, R1172, R1173, R1174, R1175, R1176, R1177, R1178, R1179, R1180, R1181, R1182, R1183, R1184, R1185, R1186, R1187, R1188, R1189, R1190, R1191, R1192, R1193, R1194, R1195, R1196, R1197, R1198, R1199, R1200, R1201, R1202, R1203, R1204, R1205, R1206, R1207, R1208, R1209, R1210, R1211, R1212, R1213, R1214, R1215, R1216, R1217, R1218, R1219, R1220, R1221, R1222, R1223, R1224, R1225, R1226, R1227, R1228, R1229, R1230, R1231, R1232, R1233, R1234, R1235, R1236, R1237, R1238, R1239, R1240, R1241, R1242, R1243, R1244, R1245, R1246, R1247, R1248, R1249, R1250, R1251, R1252, R1253, R1254, R1255, R1256, R1257, R1258, R1259, R1260, R1261, R1262, R1263, R1264, R1265, R1266, R1267, R1268, R1269, R1270, R1271, R1272, R1273, R1274, R1275, R1276, R1277, R1278, R1279, R1280, R1281, R1282, R1283, R1284, R1285, R1286, R1287, R1288, R1289, R1290, R1291, R1292, R1293, R1294, R1295, R1296, R1297, R1298, R1299, R1300, R1301, R1302, R1303, R1304, R1305, R1306, R1307, R1308, R1309, R1310, R1311, R1312, R1313, R1314, R1315, R1316, R1317, R1318, R1319, R1320, R1321, R1322, R1323, R1324, R1325, R1326, R1327, R1328, R1329, R1330, R1331, R1332, R1333, R1334, R1335, R1336, R1337, R1338, R1339, R1340, R1341, R1342, R1343, R1344, R1345, R1346, R1347, R1348, R1349, R1350, R1351, R1352, R1353, R1354, R1355, R1356, R1357, R1358, R1359, R1360, R1361, R1362, R1363, R1364, R1365, R1366, R1367, R1368, R1369, R1370, R1371, R1372, R1373, R1374, R1375, R1376, R1377, R1378, R1379, R1380, R1381, R1382, R1383, R1384, R1385, R1386, R1387, R1388, R1

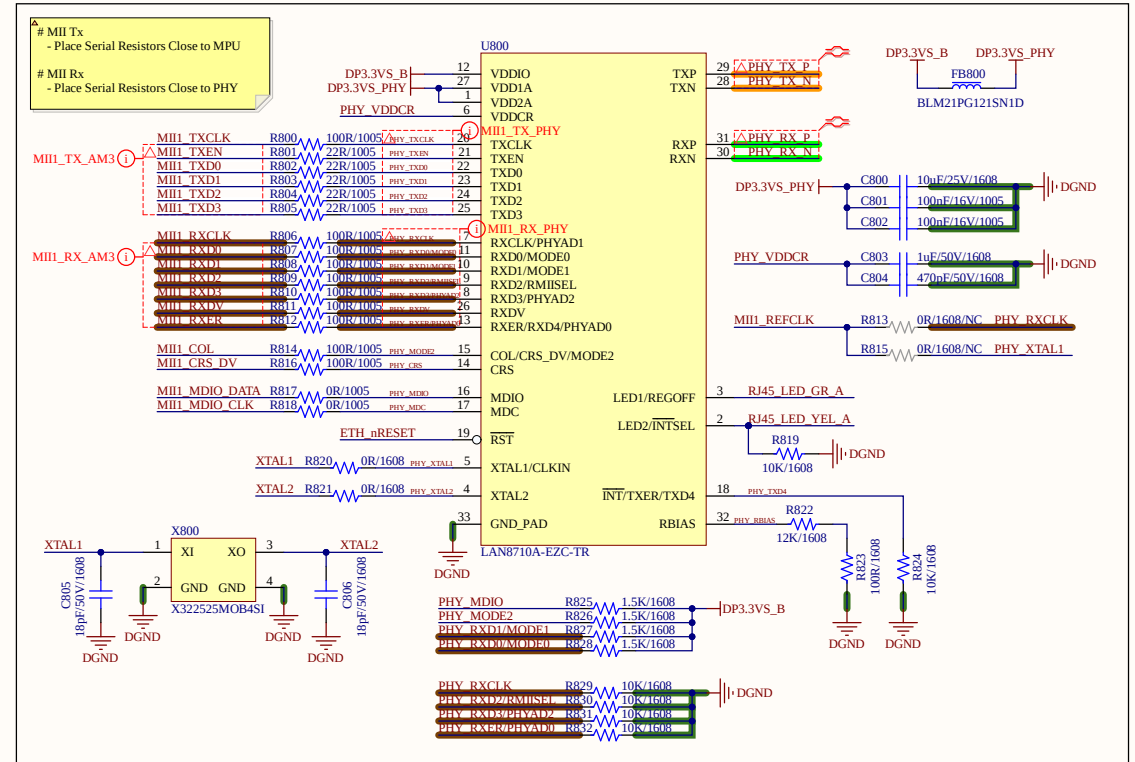
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A6	NC6	NG67	I7
A7	NC7	NG68	I8
A8	NC8	NG69	I9
A9	NC9	NC70	K1
B1	NC10	NC71	K2
B2	NC11	NC72	K3
B3	NC12	NC73	K4
B4	NC13	NC74	K5
B5	NC14	NC75	K6
B6	NC15	NC76	K7
B7	NC16	NC77	K8
B8	NC17	NC78	K9
B9	NC18	NC79	M1
B10	NC19	NC80	M2
B11	NC20	NC81	M3
B12	NC21	NC82	M4
B13	NC22	NC83	M5
B14	NC23	NC84	M6
B15	NC24	NC85	M7
B16	NC25	NC86	M8
B17	NC26	NC87	M9
B18	NC27	NC88	M10
B19	NC28	NC89	M11
B20	NC29	NC90	M12
B21	NC30	NC91	N1
B22	NC31	NC92	N2
B23	NC32	NC93	N3
B24	NC33	NC94	N4
B25	NC34	NC95	N5
B26	NC35	NC96	N6
B27	NC36	NC97	N7
B28	NC37	NC98	N8
B29	NC38	NC99	N9
B30	NC39	NC100	N10
B31	NC40	NC101	N11
B32	NC41	NC102	N12
B33	NC42	NC103	N13
B34	NC43	NC104	N14
B35	NC44	NC105	N15
B36	NC45	NC106	N16
B37	NC46	NC107	N17
B38	NC47	NC108	N18
B39	NC48	NC109	N19
B40	NC49	VSF1	ER
B41	NC50	VSF2	F1
B42	NC51	VSF3	F2
B43	NC52	VSF4	F3
B44	NC53	G10	F4
B45	NC54	VSF5	F5
B46	NC55	VSF6	F6
B47	NC56	VSF7	F7
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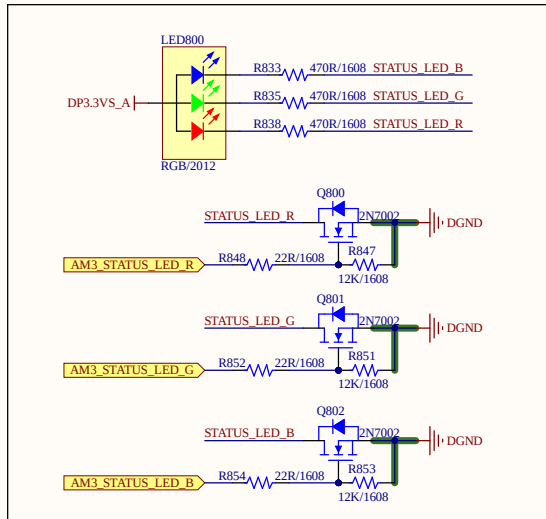
[8] Sitara I/O Bank 4/5



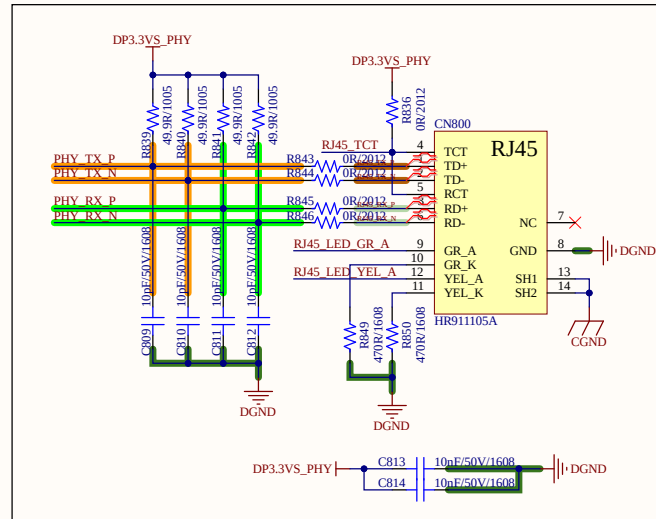
Ethernet PHY (10/100Mbps)



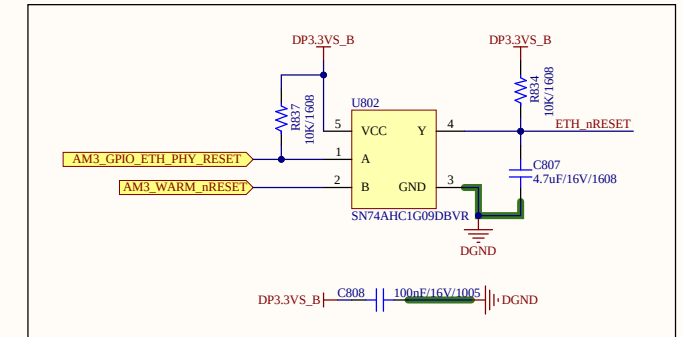
User LEDs



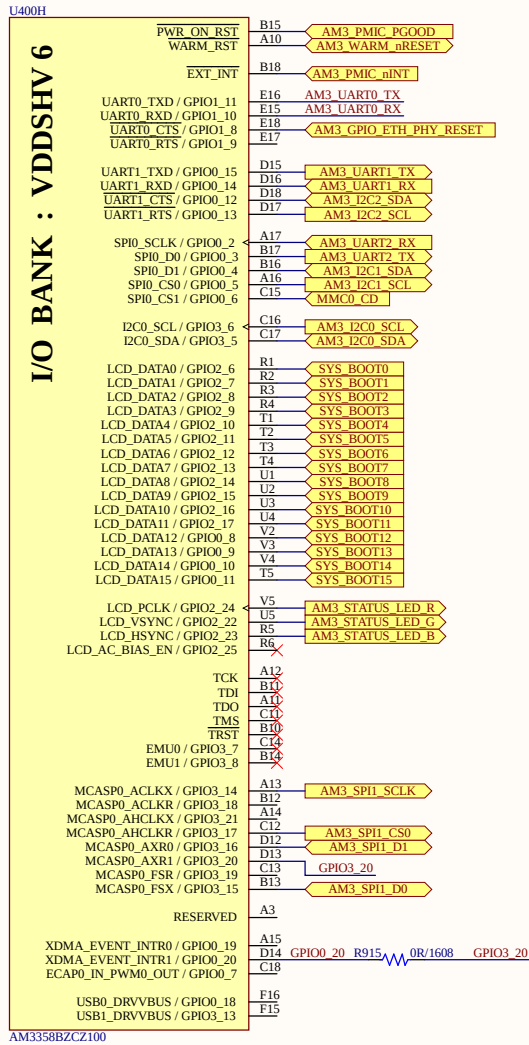
RJ45 Connector



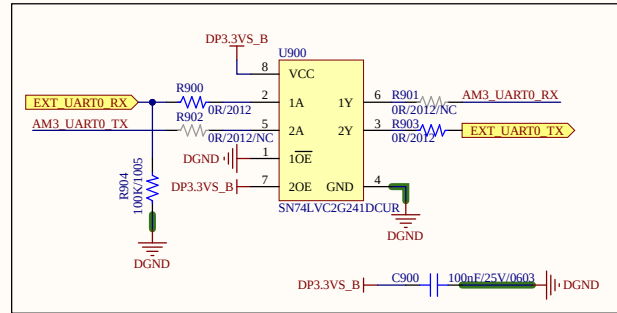
Ethernet PHY Reset



[9] Sitara I/O Bank 6

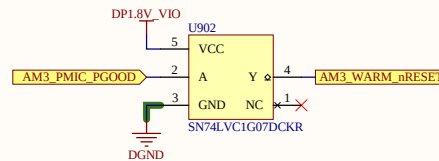


UART0 External Connector

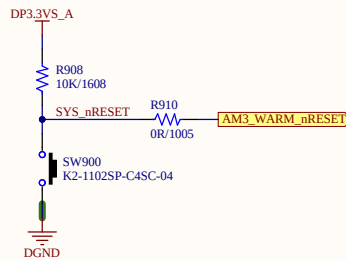


System Reset

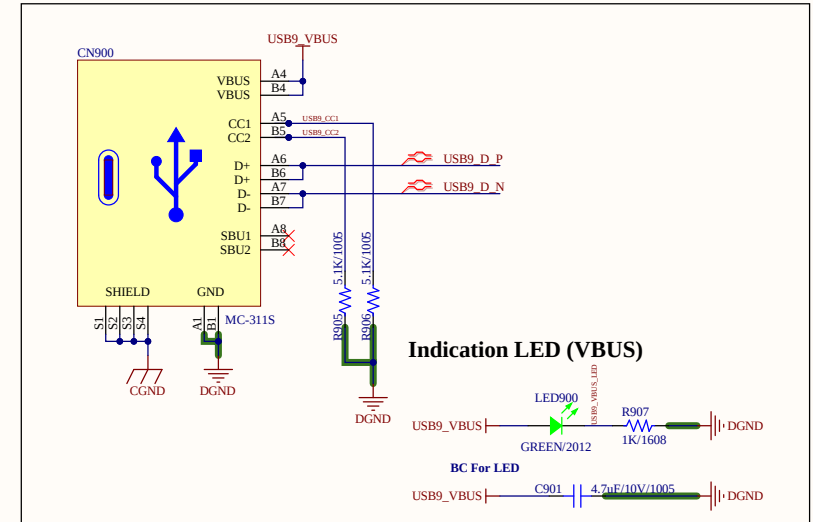
Power-On Reset



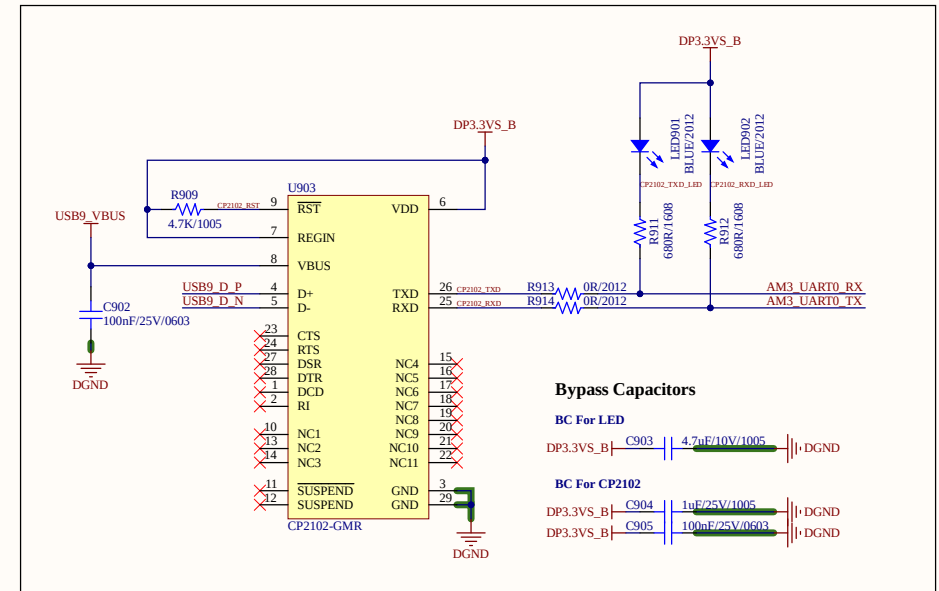
Reset Switch



USB Connector (Type-C)

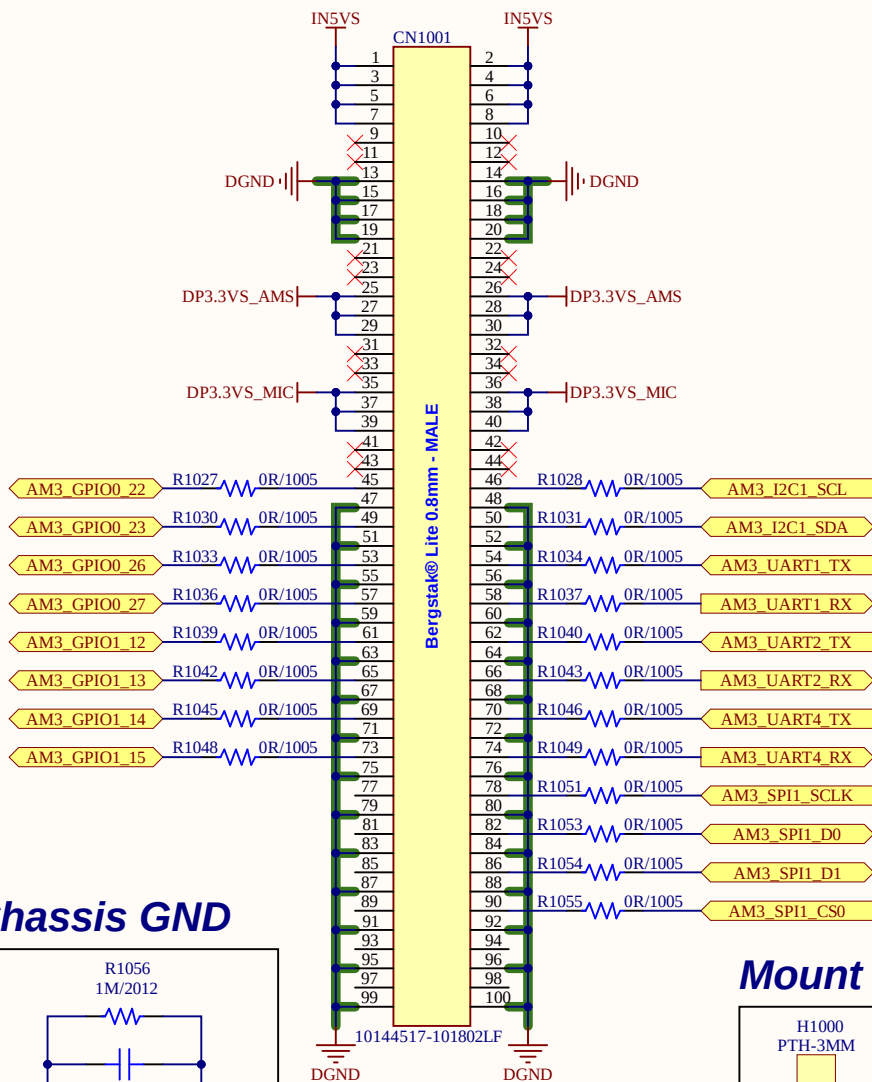


USB - UART Bridge

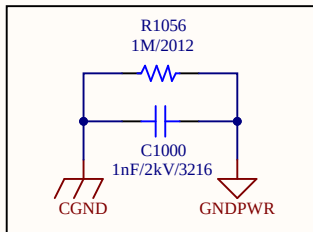


Title		Rev
Sitara_Project_Rev_A.PrjPcb		*
Doc		Sitara_IO_Bank_6.SchDoc
Sheet 9 of 10	Author ..	
Date 2026-01-31		
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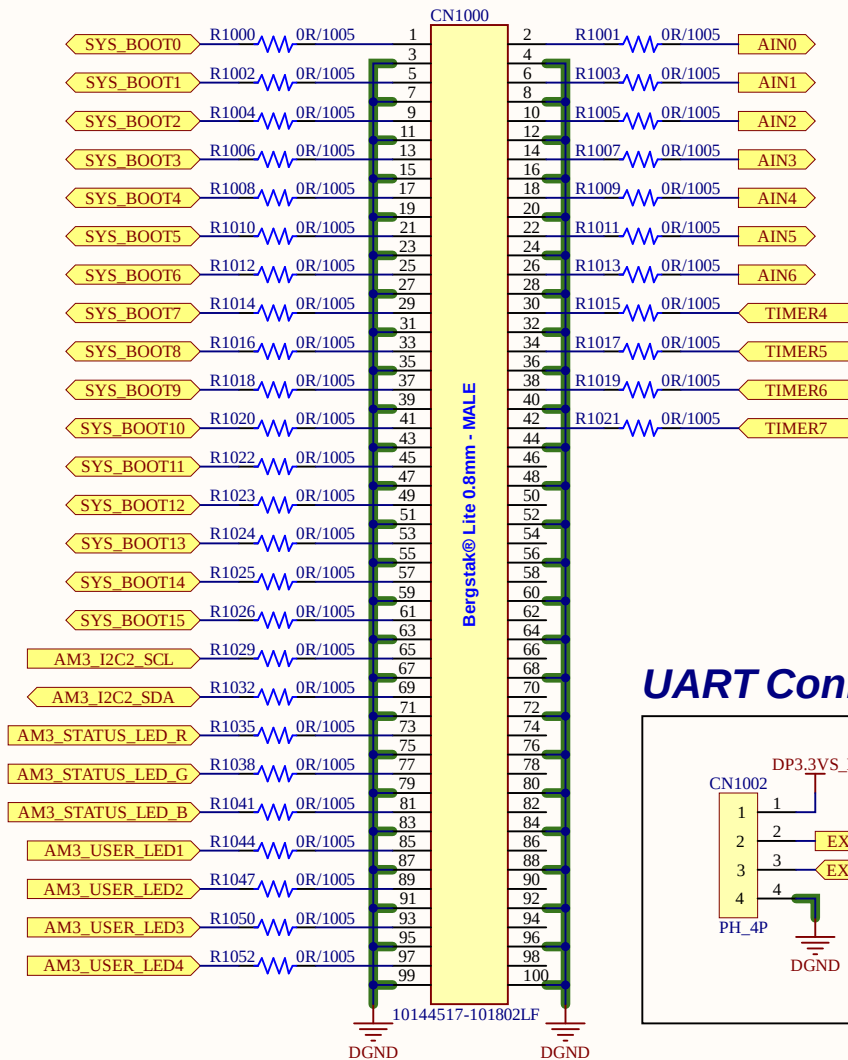
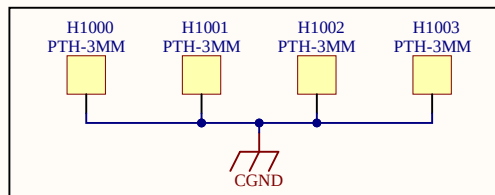
[10] Connectors



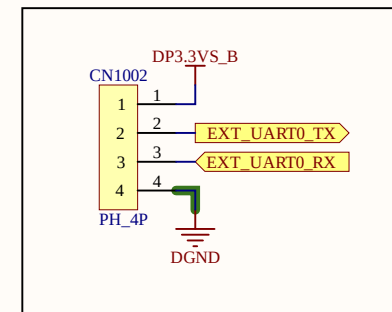
Chassis GND



Mount Holes



UART Connector



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